

Chapter 20 Study Guide

Key Concepts

20.1 The Meaning of Oxidation and Reduction

- Oxidation and reduction always occur simultaneously. The substance gaining oxygen is oxidized, while the substance losing oxygen is reduced.
- Losing electrons is oxidation. Gaining electrons is reduction.
- Iron, often used in the form of the alloy steel, corrodes by being oxidized to ions of iron by oxygen.

20.2 Oxidation Numbers

- As a general rule, a bonded atom's oxidation number is the charge that it would have if the electrons in the bond were assigned to the atom of the more electronegative element.
- An increase in the oxidation number of an atom or ion indicates oxidation. A decrease in the oxidation number of an atom or ion indicates reduction.

20.3 Balancing Redox Equations

- If the oxidation number of an element in reacting species changes, then that element has undergone either oxidation or reduction. Therefore, the reaction as a whole must be a redox reaction.
- In a balanced redox equation, the total increase in oxidation number of the species oxidized must be balanced by the total decrease in the oxidation number of the species reduced.
- To balance a redox reaction using half-reactions, write separate half-reactions for the oxidation and the reduction. After you balance atoms in each half-reaction, balance electrons gained in the reduction with electrons lost in the oxidation.

Vocabulary

- | | |
|---|---|
| ● half-reaction (p. 650) | ● oxidation-reduction reaction (p. 632) |
| ● half-reaction method (p. 650) | ● oxidizing agent (p. 634) |
| ● oxidation (p. 633) | ● redox reaction (p. 632) |
| ● oxidation number (p. 639) | ● reducing agent (p. 634) |
| ● oxidation-number-change method (p. 647) | ● reduction (p. 633) |